

Skill-Connect Ethiopia — Project Documentation & Hackathon Playbook

Event: Cursor AI Hackathon Ethiopia — July 2026 **Venue:** Adwa Victory Memorial Park, Addis Ababa **Tagline:** *The Human Capability Acceleration Engine for the AI Era*

0. How to Use This Document

This is your single source of truth for the next 48 hours. It answers three questions at every stage: **what are we building, in what order, and who does what**. If a decision isn't in here, make the simplest choice that keeps the demo working — don't gold-plate.

Print or pin the **Section 10 Hour-by-Hour Plan** — that's the one you'll actually look at every few hours.

0.5 Do This Before You Write Any Code

Ethio Telecom is judging, may extend internship/hire offers to standout participants, and can grant you a demo Telebirr API. Investors will also walk the floor. None of this costs build time if you handle it now:

1. **Register at developer.ethiotelecom.et** — use the **same email address you registered with on Luma**. Do this tonight, not tomorrow morning; account approval may take time.
 2. Once registered, request **demo/sandbox Telebirr API credentials** — this is what upgrades your payment flow from a mock to a real (sandbox) integration, which is exactly what Ethio Telecom's judges will want to see.
 3. Assign **one teammate** to own this relationship: check email/dashboard for credential approval periodically through Day 1, so integration isn't blocked waiting on someone.
 4. Prep a **60-second elevator pitch** and a **2-minute technical deep-dive** (Section 12) — you will give both multiple times, informally, to people who just walk up.
-

1. Executive Summary & Core Philosophy

Skill-Connect Ethiopia is an AI-powered workforce infrastructure platform that prepares Ethiopian youth for an AI-driven job market. It goes beyond job matching — it's a personal AI career coach that surfaces latent human talent, builds interactive learning roadmaps, verifies real ability through scored practice, and connects capability-verified talent directly with Small and Medium Enterprises (SMEs).

The Problem: AI is automating entry-level roles faster than curricula can adapt. Students are learning outdated skills and feel anxious about their futures. Local SMEs want to adopt AI but can't find fluent, verified talent — resumes and self-reported skills don't tell them who can actually do the job — and every time an experienced employee leaves, that SME loses know-how it never wrote down.

Our Philosophy ("Co-Intelligence"): AI will not replace Ethiopian youth. A human walking *with* AI will replace a human working alone. Skill-Connect helps young people thrive by treating AI as a capability multiplier — one that trains them, scores them honestly, and helps SMEs keep growing even as their teams change, always keeping a human at the center of that growth.

Why this wins a hackathon: judges reward (1) a sharp, local, well-defined problem, (2) a working end-to-end demo — not slides, (3) visible AI orchestration (they're judging *Cursor* usage too), (4) a believable path to revenue, (5) real local-infrastructure integration, and (6) a mechanism that's actually defensible on privacy and security grounds. Sections 7, 9, 12, and 13 are built specifically to hit all six.

2. The Core Mechanism — How Matching Actually Works

This is the engine behind every pillar below. Understand this section and you can explain the whole product in one breath.

1. **Sandbox challenges produce a real, graded score — not a self-report.** Every challenge a job-seeker completes in the Sandbox (Pillar 2) is evaluated by AI against a rubric, producing a numeric **Sandbox Score** per competency (e.g. *"Prompt Engineering: 82/100, Task Accuracy: 74/100, Communication: 88/100"*). This is what makes the platform credible to companies: verified performance, not a resume claim.
2. **Companies define reusable Role Skill Templates**, not one-off job posts. An SME builds a checklist once — *"Retail Inventory Assistant needs: Excel basics ≥70, AI prompt literacy ≥75, customer comms ≥65"* — and saves it. Any candidate whose Sandbox Scores clear that bar becomes eligible for *any* future opening that reuses the template, not just a single posting.

3. **Notifications are template-triggered, not a surveillance feed.** When a candidate's score crosses a saved template's threshold, the SME is notified: *"3 candidates now qualify for your Inventory Assistant template."* Companies only ever see activity tied to a template they explicitly built (see Section 9 for why this matters for privacy).
 4. **A resigning employee's handover becomes a real, custom sandbox challenge.** Instead of only producing a text brief, the AI turns the outgoing employee's described day-to-day into an actual scored challenge — reusing the same challenge-generation logic Pillar 2 already needs, fed with real job content instead of a generic template. Incoming candidates get scored against *that specific job*, not a generic rubric. The same challenge can be reused in "training mode" once someone's hired, doubling as onboarding practice.
 5. **Optional "Workplace Pressure Simulation."** Certain sandbox challenges can be run with a strict, demanding AI "manager" persona — tight deadlines, blunt feedback on the *work*, ambiguous asks, the way a real first week can feel. This is opt-in, never default, and is bounded by the rule in Section 9: the AI is critical of output, never demeaning of the person, with a visible toggle back to supportive coaching mode at any time.
-

3. Product Architecture — Four Pillars

Pillar 1: AI Talent Discovery Coach (Job-Seeker View)

Conversational AI career coach (text, voice optional) that replaces static multiple-choice assessments — and doubles as the platform's general assistant (see box below).

- Situational intake interview in **Amharic or English**, understood as **plain natural conversation**, not rigid multiple choice
- Surfaces latent human skills (adaptability, problem-solving, empathy) alongside technical/AI skills
- Parses conversation into a structured JSON skill matrix in real time
- Renders a live animated **"Skill Radar"** chart
- Recommends 2–3 AI-resilient career paths, with the specific courses/challenges in the Sandbox that will get their CV and scores ready for that path

One engine, two jobs: the same conversational AI model and prompt-handling logic that powers the intake interview also powers a small **"Ask Skill-Connect"** help bubble available site-wide — a plain-English chatbot a user or SME can ask things like *"how does matching work?"* or *"how do I post a transition role?"* Reusing the same engine keeps this from becoming a second build.

Pillar 2: The "Walk-With-AI" Sandbox (Learning + Scoring View)

Interactive learning roadmap that both teaches and **certifies**, not a static course list.

- Node-tree UI (built with `@xyflow/react`) generated dynamically from the user's skill matrix and, where relevant, target role
- Each node = a localized Ethiopian business micro-challenge (e.g., "Help a Merkato textile shop write a WhatsApp catalog with AI"), graded by AI against a visible rubric on completion
- User solves it *with* an embedded AI assistant — they learn to direct and prompt, not to code
- Completing a node issues a **verified capability badge + Sandbox Score**, both stored on the profile
- Some nodes can be run in **Workplace Pressure Simulation** mode (Section 2, point 5) for candidates who want to practice under real deadline pressure before an actual interview

Pillar 3: SME Talent Matcher + Company Profiles (Employer View)

- **Company Profiles:** every SME gets a public profile page — logo, industry, size, a short "AI-adoption journey" blurb, and a track record of roles filled through the platform.
- **Role Skill Templates (Section 2, point 2):** SMEs build a reusable scoring checklist per role type instead of writing a fresh job post every time.
- SMEs can also just type a plain-language problem — *"I need someone to automate my retail inventory"* — and the system proposes a template automatically via semantic search (pgvector) against candidate skill matrices and scores.
- Returns a ranked list with a **Match Score (0–100%)** and a **Gap Analysis** ("candidate is strong in prompt engineering, would need onboarding in inventory tools")
- **Notifications** fire automatically when new candidates clear a saved template's thresholds
- **1-click shortlist/hire action**

Pillar 3b: Institutional Handover Mode

When an employee resigns, there's normally a scramble and lost knowledge. Skill-Connect turns that transition into a structured, AI-assisted, *scored* handover:

1. SME flags a posting as a **Transition Role**.
2. If willing, the outgoing employee goes through a short **AI Handover Interview** — reusing the Pillar 1 engine — describing recurring tasks, tools, shortcuts, and who they coordinate with. They get a chance to **review and redact** the result before anything is saved (no client names or internals leak by accident).
3. The AI produces two things: a readable **Continuity Brief** for onboarding, and a **custom Sandbox challenge** modeled on that specific job (Section 2, point 4).

4. Incoming candidates are scored against that real challenge, so matching reflects *this exact departure*, not a generic rubric.
5. Once hired, the new employee gets the Continuity Brief plus the same challenge in "training mode" for onboarding practice.

Why it's worth the scope: it reuses Pillar 1's engine and Pillar 2's challenge-scoring logic almost as-is (Section 8), so the build cost is small, but it's the strongest answer to "how does this help real companies, not just individuals" — exactly what a technical judge will ask.

Pillar 4: Admin Console

A judge-visible surface most hackathon teams skip — and it doubles as your live-demo insurance policy (details in Section 4).

4. User Roles & Access

Role	How they sign up	What they see
<code>job_seeker</code>	Self-serve signup (email or phone)	Pillars 1 & 2, their own profile/badges/scores
<code>sme</code>	Self-serve signup, marked <code>pending_verification</code> initially	Pillar 3, their company profile, their templates, their postings
<code>admin</code>	Not self-serve. Seeded directly in Supabase via SQL, or promoted through a protected <code>/admin/promote</code> route gated by a hardcoded invite code (hackathon-only shortcut)	Pillar 4 — full platform view

Admin access — exact steps

1. In Supabase Studio → Table Editor → `profiles`, manually set one seed user's `role` to `admin`. Fastest, most reliable method for a 48-hour build.
2. Admin logs in through the same `/login` page as everyone else. The app checks `role` after auth and routes `admin` users to `/admin` instead of `/dashboard`.
3. Admin console shows:

- Total job seekers, SMEs, postings, matches made, transition roles in progress (live counters, good visual for judges)
- A searchable table of all skill matrices and sandbox scores (for live debugging/demo recovery)
- A "seed demo data" button that re-populates realistic Addis Ababa test profiles, company profiles, and pre-scored candidates in one click — **this is your safety net if live AI calls fail during judging**
- Simple approve/reject toggle for SME verification (pending_verification → verified)

5. What's In It for Companies — the Free Hook

The paid matching flow only works once SMEs already trust you. So the funnel starts free, before any hiring ask:

1. **Free "AI Readiness Snapshot":** a 5-question form or 2-minute chat (same engine as Pillar 1) that returns an instant, specific report — *"Here are 3 tasks in your business an AI-fluent hire could take off your plate this month."* Costs almost nothing to build, positions you as a free consultant, not a job board.
2. **Free Continuity Brief on resignation, no posting required.** *"Someone just quit? Capture their knowledge in a 10-minute AI interview, free."* This is an acute, immediate pain, not a someday-maybe need — it gets SMEs into the system at their moment of highest urgency. The natural next step is "now let AI find your replacement," which is where paid matching begins.
3. **Company Profile as a status symbol:** verified SMEs get a shareable "AI-Forward Business" badge they can post to their own Telegram/Facebook page — a vanity hook that costs nothing and spreads the platform for free.

Suggested SME funnel order: Free Readiness Snapshot → Free Handover Brief (if/when needed) → Role Skill Template + paid matching once they're already relying on you → Verified Company Profile badge as the retention/status layer.

6. Monetization Model (Investor/Judge Slide)

Stream

Model

Notes

B2B (Primary)	SMEs pay a subscription or per-hire fee to access AI-matched, verified talent once they've used the free hooks in Section 5	Bypasses traditional recruiters; core revenue driver
B2B (Add-on)	Premium tier for saved Role Skill Templates + auto-notifications, and priority Institutional Handover	Natural upsell tied to something SMEs already need
B2G & Universities	Institutional dashboards for MinT (Ministry of Innovation & Technology) and universities tracking regional digital-skill uptake	Aggregate, anonymized data only
B2C (Freemium)	Free: AI assessment, roadmap, and basic Sandbox scoring. Paid: mock interviews under Workplace Pressure Simulation, verified premium badges	Keeps youth access free — good for the "impact" narrative judges like

Payment Method — What to actually build in 48 hours

Because Ethio Telecom is judging and has offered a demo API, **Telebirr becomes your primary named rail**, with Chapa as a secondary/backup story:

1. **Follow Section 0.5** — register at `developer.ethio telecom.et` with your Luma email, request sandbox Telebirr credentials as early as possible on Day 1.
2. **If credentials arrive in time:** wire a real **Telebirr sandbox checkout** for the SME "Upgrade" flow. This is the single highest-value technical flex for the Ethio Telecom judges specifically.
3. **If credentials don't arrive in time (have this ready regardless):** fall back to a **mocked checkout UI** styled after Telebirr's real flow. Mention **Chapa** in the pitch as the multi-rail backup for cards/other mobile-money providers.
4. Say explicitly in the pitch what's real vs. mocked. Judges respect honesty about scope far more than a shaky live demo.

7. Technology Stack

- **Frontend:** Typescript, Next.js 15 (App Router), Tailwind CSS, Shadcn UI, Framer Motion (polish pass only)

- **Backend/DB:** Supabase — PostgreSQL for structured data, [pgvector](#) extension for semantic SME↔talent matching, **Row Level Security (RLS)** enforced on every sensitive table (Section 9)
 - **AI Orchestration:** Vercel AI SDK (streams AI responses directly into React components)
 - **LLM Provider:** OpenAI or Gemini API for conversational intake, handover interviews, sandbox challenge grading, and matching logic
 - **Voice (optional stretch):** ElevenLabs or Whisper for Amharic/English voice interaction — only attempt this in Phase 4 if core flows are solid
 - **Visualization:** [@xyflow/react](#) for the learning roadmap node tree; Recharts or a custom SVG for the Skill Radar
 - **Payments:** Telebirr sandbox API (primary, via Ethio Telecom developer portal), Chapa (backup/multi-rail narrative)
-

8. Data Schema (Supabase / PostgreSQL)

```
-- profiles
id          UUID PRIMARY KEY
role        ENUM ('job_seeker', 'sme', 'admin')
full_name   TEXT
bio         TEXT
phone       TEXT
region      TEXT
created_at  TIMESTAMP

-- company_profiles (one per SME)
id          UUID PRIMARY KEY
sme_id      UUID REFERENCES profiles(id)
company_name TEXT
industry    TEXT
size        TEXT
logo_url    TEXT
about       TEXT
verified    BOOLEAN DEFAULT FALSE

-- skill_matrices
id          UUID PRIMARY KEY
user_id     UUID REFERENCES profiles(id)
skills_json JSONB
readiness_score INTEGER
created_at  TIMESTAMP
```



```

-- sandbox_scores (new — one row per completed challenge)
id          UUID PRIMARY KEY
user_id     UUID REFERENCES profiles(id)
node_id     TEXT          -- which sandbox challenge was completed
scores_json JSONB          -- {"prompt_engineering": 82, "task_accuracy": 74,
"communication": 88}
mode        ENUM ('standard', 'pressure_simulation')
completed_at TIMESTAMP

-- badges
id          UUID PRIMARY KEY
user_id     UUID REFERENCES profiles(id)
node_id     TEXT
badge_name  TEXT
awarded_at  TIMESTAMP

-- role_skill_templates (new — reusable SME scoring checklists)
id          UUID PRIMARY KEY
sme_id      UUID REFERENCES profiles(id)
role_name   TEXT
thresholds_json JSONB      -- {"excel_basics": 70, "ai_prompt_literacy": 75,
"customer_comms": 65}
notify_on_match BOOLEAN DEFAULT TRUE
created_at  TIMESTAMP

-- sme_postings
id          UUID PRIMARY KEY
sme_id      UUID REFERENCES profiles(id)
template_id UUID REFERENCES role_skill_templates(id) -- new: links posting to a
reusable template
description  TEXT
is_transition_role BOOLEAN DEFAULT FALSE
status      ENUM ('open', 'filled')
created_at  TIMESTAMP

-- continuity_briefs (only exists for transition roles)
id          UUID PRIMARY KEY
posting_id  UUID REFERENCES sme_postings(id)
raw_interview_json JSONB
generated_brief TEXT
custom_node_id TEXT        -- new: the AI-generated sandbox challenge modeled on this
job

```

```
reviewed_by_employee BOOLEAN DEFAULT FALSE -- new: redaction/consent gate before it's
usable
created_at      TIMESTAMP
```

```
-- matches
```

```
id          UUID PRIMARY KEY
posting_id   UUID REFERENCES sme_postings(id)
candidate_id UUID REFERENCES profiles(id)
match_score  INTEGER
gap_analysis TEXT
status       ENUM ('suggested', 'shortlisted', 'hired')
```

```
-- notifications (new — template-triggered, not a general feed)
```

```
id          UUID PRIMARY KEY
sme_id       UUID REFERENCES profiles(id)
template_id  UUID REFERENCES role_skill_templates(id)
candidate_id UUID REFERENCES profiles(id)
seen         BOOLEAN DEFAULT FALSE
created_at   TIMESTAMP
```

```
-- payments
```

```
id          UUID PRIMARY KEY
sme_id       UUID REFERENCES profiles(id)
provider     ENUM ('telebirr', 'chapa', 'mock')
amount       INTEGER
status       ENUM ('pending', 'paid')
created_at   TIMESTAMP
```

Keep every table this flat. Do not add extra normalization tables under time pressure — Cursor's Composer works faster and more reliably against a small, obvious schema.

9. Trust & Security Model (Do Not Skip This)

The matching mechanism in Section 2 is powerful precisely because it's built on real scores — which is exactly why it needs real boundaries. Treat these as hard rules, not stretch goals:

1. **Companies never see raw conversation transcripts.** A candidate's intake chat or handover interview may surface personal context (family situation, disability, reason for job-hunting). SMEs only ever see *derived* outputs: Sandbox Scores, badges, and Gap Analysis text. This is enforced in the query layer, not just hidden in the UI.

2. **Opt-in discoverability, not scraping.** A candidate must toggle "open to being matched" before their scores count toward any SME's Role Skill Template results — the same pattern as LinkedIn's "open to work." Without this, the platform becomes passive surveillance of job seekers who never agreed to be evaluated by a specific employer.
 3. **Notifications are template-scoped.** An SME is only notified about candidates who clear a template *they* built and *the candidate* opted into — never a general "here's everyone's scores" feed.
 4. **Continuity Briefs are private to the issuing SME,** never public, and require the outgoing employee's review/redaction before they're usable — they may unintentionally include client names or internal specifics.
 5. **Workplace Pressure Simulation is bounded.** The AI "strict manager" persona stays critical of the *work product*, never of the *person* — no personal insults, no discouragement tied to identity. A visible toggle back to supportive coaching mode is always available. This applies to Claude/LLM prompt design directly: the system prompt for this mode must explicitly encode this boundary.
 6. **Enforced with Supabase Row Level Security (RLS), not just app logic:** job seekers can only query their own matrix/scores; SMEs can only query derived scores for candidates who opted into their specific template; admin access is scoped and logged. This is also a strong, concrete technical talking point for Ethio Telecom's judges — RLS policies are the kind of defensible architecture decision that shows real engineering judgment, not AI-generated boilerplate.
-

10. Step-by-Step Feature List (Build Order)

Build in this exact order. Each feature is demoable on its own, so if you run out of time you still have a working story.

1. **Auth + role routing** — signup/login, `job_seeker` vs `sme` vs `admin` redirect, RLS policies scaffolded from the start (Section 9)
2. **Job-seeker intake chat** — streaming AI conversation → structured skill JSON saved to DB
3. **Skill Radar visualization** — renders from `skills_json`
4. **Career recommendations** — 2–3 AI-resilient roles suggested from the matrix, linked to specific Sandbox nodes
5. **Learning Sandbox node tree + scoring** — generated from matrix, at least 3 real localized challenges, each AI-graded into `sandbox_scores` on completion
6. **Badge award on challenge completion**
7. **SME signup + Company Profile page**
8. **Free "AI Readiness Snapshot"** (Section 5) — no signup friction, immediate value
9. **Role Skill Templates** — SME builds a reusable threshold checklist per role

10. **Semantic + threshold match engine** — posting/template → ranked candidate list with Match Score + Gap Analysis, respecting opt-in (Section 9)
 11. **Notifications** — fire when a candidate clears a saved template
 12. **1-click shortlist/hire action**
 13. **Institutional Handover flow** — free AI interview for outgoing employee → Continuity Brief + AI-generated custom sandbox challenge, with employee review/redaction step
 14. **Payment flow** — Telebirr sandbox if credentials are ready, otherwise Telebirr-styled mock (Section 6)
 15. **Admin console** — live counters, seed-data button, SME verification toggle
 16. **Site-wide "Ask Skill-Connect" chatbot bubble** — thin UI wrapper over the Pillar 1 engine
 17. **Optional: Workplace Pressure Simulation toggle** on select sandbox nodes, with the bounded system prompt from Section 9
 18. **Polish pass** — dark mode, Framer Motion transitions, seeded Addis Ababa test data, Amharic labels where feasible
-

11. Where the Demo Data Comes From

You cannot rely on real user sign-ups by tomorrow, and you should never use real people's names, photos, or company identities without consent. Practice and demo data should be **entirely synthetic, prepared in advance**, not generated live on stage:

1. **Generate personas ahead of time with the LLM itself.** Spend 15–20 minutes prompting your LLM provider directly (outside the app) for ~20 realistic Ethiopian job-seeker personas — name, background, short story — across sectors like retail, textiles, logistics, and agritech. Save as a static JSON seed file, and pre-assign each one plausible `sandbox_scores`.
2. **Do the same for SMEs:** 8–10 fictional company profiles modeled on common Addis Ababa small-business sectors, each with at least one saved Role Skill Template. Use invented company names — never real, identifiable businesses (and never Ethio Telecom's name/brand for fictional demo data).
3. **Ground it in real context, not real records.** You can reference general, publicly known facts about Ethiopia's labor market and youth unemployment trends to make personas believable — don't fabricate specific statistics and present them as sourced data in the pitch.
4. **Load this static seed file via the Admin Console's "seed demo data" button** so rehearsal and stage always get the same reliable dataset in one click.
5. **Practice on this exact seed data** for at least your last 10 rehearsal runs, so the live conversational parts (Pillar 1 intake, the SME query, the notification firing) are ones you've typed before and know the expected output for.

13. The 2-Minute Demo Script

Judges remember narratives, not feature lists. Run it in this order:

1. **(10s) The hook:** "Meet Selam, a university grad in Addis. She's anxious — she doesn't know if her skills matter in an AI economy." Open the app as a fresh job-seeker.
2. **(25s) Pillar 1 live:** Short real conversation with the AI coach (pre-planned answers). Skill Radar animates in.
3. **(25s) Pillar 2 live:** Solve one sandbox challenge with the embedded AI — show the graded score appear, badge pops.
4. **(20s) Free hook, SME side:** Switch to an SME account, run the free AI Readiness Snapshot in a few seconds — "here's what AI could take off your plate."
5. **(25s) Pillar 3 live:** Show the Company Profile, then the SME's saved Role Skill Template — a notification fires live showing Selam just qualified.
6. **(15s) Pillar 3b:** "This posting is actually a transition role — someone resigned last week." Show the AI-generated Continuity Brief and the custom sandbox challenge modeled on that exact job.
7. **(10s) Payment + Admin:** SME clicks "Upgrade" → Telebirr checkout → Admin console's live counters tick up.
8. **(10s) Close:** One sentence on the business model (Section 6) and the local-first thesis: *"Built for Ethiopia's SMEs and youth, in Amharic, on a payment rail Ethiopians already trust — and scored, not guessed."*

Fallback rule: if any live AI call is slow or fails during judging, hit the admin "seed demo data" button and continue narrating over pre-seeded data without breaking stride. Rehearse this fallback too.

14. Judges & Investor Engagement (Read This Before the Floor Opens)

Ethio Telecom judges will focus on **technical depth**, and standout participants may be shortlisted for internships or hiring conversations. Investors and other organizations will also walk the floor throughout the event. Treat both as first-class parts of the hackathon, not a distraction from coding:

- **Everyone on the team should be able to explain the architecture**, not just one person. Take 10 minutes in the polish phase to walk each other through the parts you

didn't personally build — especially the RLS/security model in Section 9, since it's a strong, specific thing to discuss.

- **Have two pitches ready:**
 - A **60-second elevator pitch** (hook + problem + the four pillars in one sentence each) for anyone who casually stops by.
 - A **2–3 minute technical deep-dive** for Ethio Telecom specifically: open Cursor and show the Vercel AI SDK streaming route, the pgvector match query, the RLS policies, and the Telebirr integration code directly.
 - **Actively look for a natural moment to mention Telebirr integration** to any Ethio Telecom representative who visits.
 - **Don't oversell what's mocked.** Say plainly what's live versus mocked. This is a strength in front of technical judges, not a weakness.
 - Assign **one teammate as the designated first responder** when someone new walks up, but rotate who gives the technical walkthrough so every teammate gets facetime.
-

15. What Makes Judges Say "This One's Different"

- **Verified, not self-reported.** Sandbox Scores turn "I know AI tools" into a graded number a company can trust — this is the single strongest technical/product differentiator in the whole project.
 - **Localization is real, not decorative** — Amharic intake, Addis Ababa test data, and (if credentials arrive) a live Telebirr sandbox integration.
 - **You show AI orchestration, not just AI output** — the Sandbox literally teaches and grades "directing AI," mirroring what the hackathon itself is testing.
 - **Institutional Handover Mode** tells judges you thought about a real, unglamorous business problem instead of only the flashy parts.
 - **A real security model (Section 9)**, not an afterthought — opt-in matching, scoped notifications, RLS enforcement. This is rare in hackathon projects and will stand out specifically to technical judges.
 - **A monetization story with three real, distinct buyers**, a free-to-paid SME funnel (Section 5), and a natural upsell — most hackathon projects only have one revenue idea.
 - **Honesty about scope** reads as senior engineering judgment, not a weakness.
-

16. Risk Register — Plan for These Before They Happen

Risk

Mitigation

LLM API rate-limited or slow during judging	Admin "seed demo data" button; pre-recorded 30s backup clip of the AI coach flow on a phone
Telebirr sandbox credentials don't arrive in time	Ready-made Telebirr-styled mock checkout (Section 6) as a same-day fallback
Venue WiFi fails	Test the app on mobile hotspot the night before; confirm Supabase and your LLM provider are reachable over it
Semantic/threshold match returns nonsense on stage	Pre-test the exact SME query and template you'll demo live, at least 5 times beforehand
RLS misconfigured, exposing candidate data across roles	Test explicitly as both a job_seeker and sme account before demo freeze — this is a security bug, not just a display bug
Workplace Pressure Simulation reads as mean rather than "tough but fair"	Keep it opt-in, test the tone yourself first, cut it from the demo entirely if it doesn't land right in rehearsal
Amharic text rendering breaks on a judge's browser	Confirm font stack includes a Unicode Ethiopic-script font (e.g., Noto Sans Ethiopic) and test on the actual demo machine
Running out of time before Pillar 3b or payments	Section 10's build order is deliberate — Pillars 1, 2, and core Pillar 3 alone are still a coherent, demoable story

17. Suggested Team Split (adjust to your team size)

- **1 person:** Supabase schema, auth, RLS policies, admin console
- **1–2 people:** Pillar 1 (AI coach + Skill Radar) + site-wide chatbot bubble
- **1–2 people:** Pillar 2 (Sandbox node tree + AI grading + challenges)
- **1 person:** Pillar 3 (Company Profiles + Role Skill Templates + matching + notifications) + Pillar 3b (Handover flow) + Telebirr integration
- **Everyone, last 4 hours:** seed data, polish, demo rehearsal, and make sure every teammate can explain at least one other person's part of the stack for the judges

End of document. Good luck — build in the order above, protect your demo with the admin fallback, register on the Ethio Telecom developer portal tonight, take the security model in Section 9 seriously, and don't touch real financial infrastructure beyond the sandbox you're given.